Practice Problems for Midterm 2

1. (Hotel questionnaire) A hotel wants to determine average customer satisfaction using a questionnaire that asks customers for a discrete score between 1 (worst) and 5 (best). 20% of the customers are retired. The average score given by the retired customers is 2, and they answer the questionnaire with probability 0.9. The average score given by the rest of the customers is 4, and they answer the questionnaire with probability 0.25. We define a Bernoulli random variable $\tilde{r}$ to represent whether a customer is retired ($\tilde{r} = 1$) or not ($\tilde{r} = 0$), another Bernoulli random variable $\tilde{q}$ to represent whether they answer the questionnaire ($\tilde{q} = 1$) or not ($\tilde{q} = 0$), and a discrete random variable $\tilde{s}$ to represent the score. We assume that $\tilde{q}$ and $\tilde{s}$ are conditionally independent given $\tilde{r}$.

   a) What is the mean of the score $\tilde{s}$?

   b) What is the mean of the *observed* score, i.e. the conditional mean of $\tilde{s}$ given $\tilde{q} = 1$?

2. (Faulty database) An entry in a database can be modeled as a Gaussian random variable $\tilde{a}$ with zero mean and variance σ^2. Due to a programming error, for a certain fixed constant x, the entry is replaced by $-x$ with probability 0.1 and x with probability 0.1, independently of the content of the entry. Otherwise, the entry remains the same. We model the resulting entry as a random variable $\tilde{b}$.

   a) Compute the variance of $\tilde{b}$.

   b) Derive the minimum mean-squared error estimator of $\tilde{a}$ given $\tilde{b} = b$, for all possible values of b.

3. (Weight-loss supplement) In order to evaluate the effectiveness of a weight-loss supplement, we record the weight gain/loss in kilograms after one month of a treatment group that takes the supplement, and a control group that doesn't. In bold, we highlight the participants who report that they exercise daily.

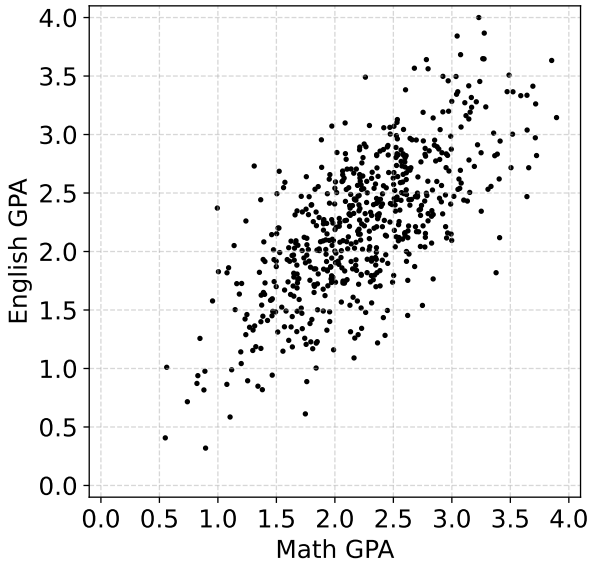
   Supplement: 0.5, **-4.6**, **-7.5**, 1.1, **-3.4**, **-1.4**, **3.2**, -1.9, **-3.6**, 2.4, 1.2, **4.2**, **-2.5**, **-3.5**, 4.7
   No supplement: 1.6, **-6.7**, 2.3, **-4.5**, **3.4**, -0.6, 2.3, 1.5, 1.7, 2.6, 0.5, 1.2, **-3.6**, 2.1, -1.2, 0.4

   a) Compute the observed average treatment effect (ATE) given by the difference in average weight gain/loss between the treatment and control groups.

   b) Estimate the ATE making any adjustments that you think are reasonable. Explain under what assumptions your estimate approximates the true ATE.

4. (Three random variables) If a random variable $\tilde{w}$ is positively correlated with another random variable $\tilde{y}$, and $\tilde{y}$ is positively correlated with a third random variable $\tilde{z}$, can $\tilde{w}$ and $\tilde{z}$ be negatively correlated? Provide an example of three such variables, or prove that it is impossible.

5. (Berkson's paradox) The following plot shows the GPA in English and Math of the students in a high school. Students whose total GPA (English + Math) is 6.5 or above are considered honors students.



a) Are English and Math GPAs uncorrelated, positively correlated or negatively correlated?

b) Conditioned on being an honors student, are English and Math GPAs uncorrelated, positively correlated or negatively correlated?

6. (Noisy measurement) We are interested in measuring a certain quantity modeled by a random variable $\tilde{x}$ with zero mean and unit variance. Our available measurements $\tilde{y} := \tilde{x} + \tilde{z}$ are corrupted by additive random noise, modeled as a random variable $\tilde{z}$ with zero mean and variance σ^2, which is independent from $\tilde{x}$.

   a) What is the best linear estimator of $\tilde{x}$ given $\tilde{y} = y$?

   b) What is the corresponding mean squared error?

   c) What happens to the estimator and the error when $\sigma \to 0$? Explain why this makes sense.

   d) What happens to the estimator and the error when $\sigma \to \infty$? Explain why this makes sense.

7. (Bernoulli random variables) Does uncorrelation imply independence for Bernoulli random variables?

8. (Exam) The following table shows the number of hours that a group of students dedicated to study for an exam, and their resulting grades.

Hours	5	6	2	6	5	5	3	3	6	8
Grade	85	90	30	80	90	80	60	40	90	85

   a) Approximate the minimum MSE estimator of the grade given the number of hours from the data and sketch it on a graph. What is the main advantage and disadvantage of this estimator?

b) Approximate the linear minimum MSE estimator of the grade given the number of hours from the data, and sketch it superposed onto the minimum MSE estimator. What is the main advantage and disadvantage of this estimator?

9. (Online poll) In online polls, young people are often overrepresented. In this problem we show how to correct for this. Our goal is to analyze an online poll for a fictitious election where there are two candidates, a Democrat and a Republican. The poll consists of n_1 young people and n_2 old people. We denote by y and o the number of young and old Democrat voters in the poll, respectively. We assume that the fraction of young voters in the population, denoted by α, is known.

   a) Derive an estimator of the proportion of Democrat voters in the election, as a function of y, o, n_1 and n_2. As a hint, it may be useful to consider the population parameters θ_{young} and θ_{old}, representing the proportion of Democrats among young and old voters in the population, respectively.

   b) Apply your estimator to a poll where 60 out of 100 participants intend to vote for the Democratic candidate. 70 out of the 100 participants are young, and 50 of these 70 are Democratic voters. The fraction of young voters in the overall population is 25%.

   c) Under what assumptions is your estimator unbiased? Justify your answer mathematically.

   d) Show that your estimator is consistent as $n_1 \to \infty$ and $n_2 \to \infty$.

10. (Herding) *Herding* is the act of manipulating poll results, so that they are more similar to other polls. Let $\tilde{m}_1$, $\tilde{m}_2$ and $\tilde{m}_3$ be random variables representing the sample proportion of voters supporting a certain candidate in three polls. The polls are independent and have the same standard error s. Poll 1 is released first. After seeing it, the second pollster uses its result to perform herding via averaging, obtaining the *herded* estimator,

$$\tilde{h}_2 := \frac{\tilde{m}_1 + \tilde{m}_2}{2}. \tag{1}$$

After Poll 2 is released, the third pollster also performs herding using the average of the two first polls,

$$\tilde{h}_3 := \frac{1}{2}\left(\frac{\tilde{m}_1 + \tilde{h}_2}{2} + \tilde{m}_3\right). \tag{2}$$

   a) Assuming $\tilde{m}_1$, $\tilde{m}_2$ and $\tilde{m}_3$ are unbiased, are $\tilde{h}_2$ and $\tilde{h}_3$ unbiased? Does herding produce bias in this situation?

   b) Derive the standard error of the herded estimators $\tilde{h}_2$ and $\tilde{h}_3$ as a function of the standard error s. Does herding improve or degrade the estimators?

   c) Derive the standard error of the estimator obtained by averaging the three non-herded estimators,

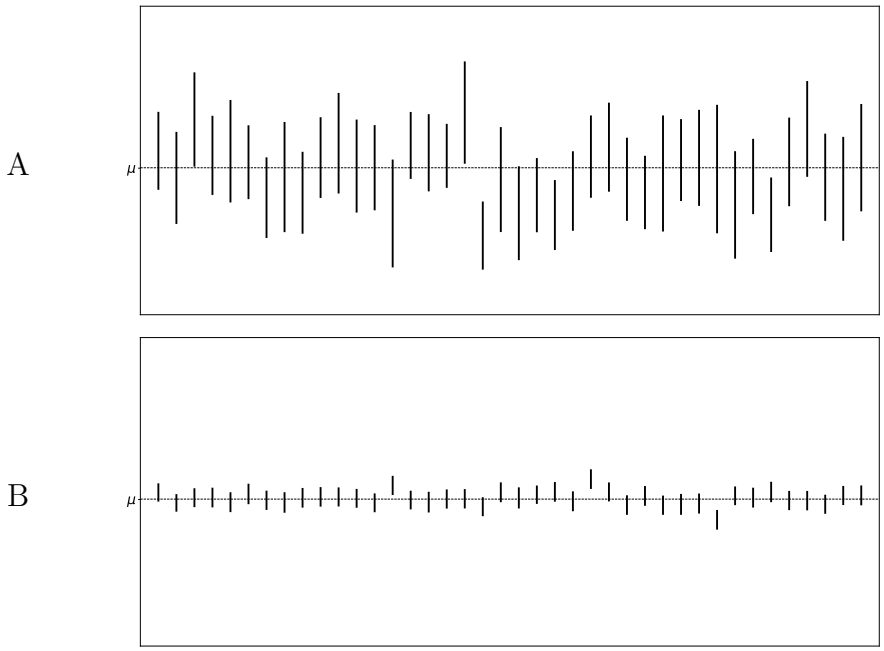
$$\tilde{m}_{\text{all}} := \frac{\tilde{m}_1 + \tilde{m}_2 + \tilde{m}_3}{3}, \tag{3}$$

and the estimator obtained by averaging the three herded estimators,

$$\tilde{h}_{\text{all}} := \frac{\tilde{m}_1 + \tilde{h}_2 + \tilde{h}_3}{3}, \tag{4}$$

as a function of the standard error s. Why is herding problematic?

11. (Confidence intervals) The following are $1 - \alpha$ confidence intervals for a certain quantity μ, estimated using the sample mean of the observed data.



a) Estimate α.

b) In each scenario, choose whether you think $X = A$ and $Y = B$, or $X = B$ and $Y = A$.

- In plot X, the confidence intervals are based on the central limit theorem. In plot Y, they are built using Chebyshev's inequality (everything else is the same).
- In plot X, the confidence intervals are built using more data than in plot Y (everything else is the same).
- In plot X, the data are sampled from a distribution that has higher variance than plot Y (everything else is the same).

12. (Cholesterol) We want to determine whether men have higher cholesterol than women in a certain population. We pick 200 men and 100 women independently at random with replacement from the population, and compute their average cholesterol levels.

a) Let σ_{men} and σ_{women} denote the standard deviations of the cholesterol level for the men and women in the population, respectively. What is the standard error of the difference between the average cholesterol levels computed from our data?

b) The average cholesterol levels we observe are 200 for the men and 180 for the women. Assuming $\sigma_{\text{men}} = \sigma_{\text{women}} = 10$, compute a 95% confidence interval for the difference between the mean cholesterol level of the men and the women in the population.

13. (Modified Gaussian bootstrap confidence interval) Bootstrap percentile confidence intervals work for estimators that are rendered approximately Gaussian by a strictly monotonic transformation, even if the transformation is not known to us. Propose an alternative approach, based on the bootstrap standard error, to build confidence intervals when the transformation is known (as in the case of Fisher's transformation).